



Department of Computer Science and Engineering

Academic Year 2021 - 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE 'A'

Course Code & Title: EC8393 & Fundamentals of Data Structures in C

Name of the Faculty member (s): Ms.P.Jothi Thilaga, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit I / One dimensional and Two dimensional arrays
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO3
- **Activity Chosen:** Minute Paper
- **Justification:**

Array concept is an important topic in C programming. Many applications can be developed using array concept in C. This activity makes the students to get a comprehensive knowledge in this concept.

- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**

At the end of the class, the students were asked to write about the array topic discussed in the class. The students expressed the understood concepts in the array topic and the content which were not clear in that particular topic. The students also write about the specific topic which is needed to be discussed or clarified in the further classes. The points on array topic shared by the students through this activity are shown in the figure 1. This activity shows whether the students can able to understand the array topic and their involvement in the particular class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PSO1
CO1	3	1	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO10	PSO1
	2	1	1	1	2	1
Justification for correlation	Needs engineering and mathematical knowledge for Array concept	Basic C programming concepts are used in analysis of engineering problems	To design solutions for complex engineering problems using array concepts	To provide complex problem solutions with relevant array operations	Able to execute and present programs using array Concepts	Able to apply array concepts in emerging trends in Information Technology

• Images / Screenshot of the practice:

Left Screenshot (11/08/2021):

Array is collection of element having a same datatype, is known as array.

Syntax:

```
char int = a[10], b[10];
```

Types of array:

- * Single dimensional array
- * Multi " "

Array: Arrays are sequence of data type having homogeneous value.

Declaration

```
data-type arr-name[arr-size];
```

Initialization

```
data-type arr-name[arr-size] = {value1, value2, value3, ...}
```

Name: K. Manoj mullikarjuna

Right Screenshot (11/08/2021):

Array: Collection of data

Array declaration Syntax: data-type arr-name [arr-size];

```
int a[10];
```

Array accessing Syntax: arr-name[index];

```
a[2] = 7
```

Array initialization Syntax:

```
data-type arr-name [arr-size] = (value1, value2, ...)
```

```
int a[10] = {5, 6, 7}
```

Array: Collection of data that holds fixed no. of. values of same type

Initializing:

```
data-type array_name [size];
```

Example

```
char greetings [6] = {'H', 'e', 'l', 'l', 'o', '\0'};
```

```
int a [2] [2] = {1, 2, 3, 4}
```

Figure 1: Minute Paper

• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- The students felt that they can able to understand the array topic clearly after conducting this activity.
- Also they can able to practice more programs on array concept.

❖ Benefit of the practice:

- Students were actively participated in this activity.
- From this activity, the muddiest parts in the concept were identified and the syntax and concepts of two dimensional arrays were explained again in the next class for well understanding.

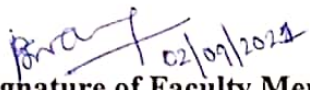
- Example programs were given and explained for their better understanding.
- Able to attend the programs using array in Internal Test.

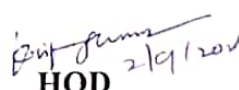
❖ **Challenges faced in implementation:**

- Some students are hesitated and lack to share the concepts that they learnt in the class.
- Make the students to know the impact and importance of sharing their views and understanding related to the topic and made them involve in the activity.

References:

- ❖ <https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html>
- ❖ <https://www.unl.edu/gradstudies/current/teaching/minute>
- ❖ <https://oncourseworkshop.com/self-awareness/one-minute-paper/>


Signature of Faculty Member


HOD